

2.10 EXERCISE

1. Prove that $\text{ROT}(\hat{X}, \theta_1)\text{ROT}(\hat{X}, \theta_2) = \text{ROT}(\hat{X}, \theta_1 + \theta_2)$
2. A frame $\{B\}$ is rotated about $\hat{X}_U$ axis of the universal coordinate system by 45 degrees and translated along $\hat{X}_U$, $\hat{Y}_U$ and $\hat{Z}_U$ by 1, 2 and 3 units, respectively. Let the position of a point Q in $\{B\}$ is given by $[3.0 \ 2.0 \ 1.0]^T$. Determine ${}^U\bar{Q}$.
3. The concept of roll, pitch and yaw angles has been used to represent the rotation of a frame $\{B\}$ with respect to the reference frame $\{U\}$, that is, ${}^U_B R$. Let us suppose that the above rotation can also be expressed by a 3×3 rotation matrix as given below.

$${}^U_B R = \begin{bmatrix} -0.250 & 0.433 & -0.866 \\ 0.433 & -0.750 & -0.500 \\ -0.866 & -0.500 & 0.000 \end{bmatrix}$$

Determine the angles of rolling, pitching and yawing.

4. A frame $[T]$ is given by

$$[T] = \begin{bmatrix} 0 & -1 & 0 & 4 \\ 0 & 0 & -1 & -5 \\ 1 & 0 & 0 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

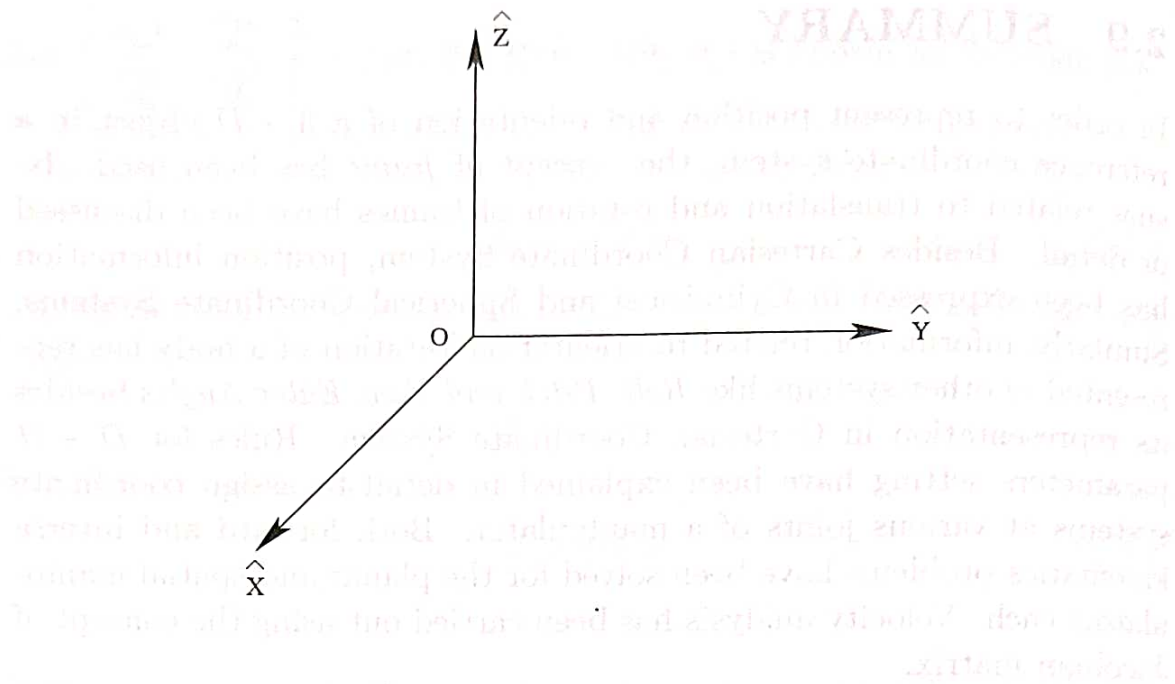


Figure 2.17: Reference frame.

- (i) Sketch the frame $[T]$ in the reference frame shown in Fig. 2.17.
 - (ii) Sketch the frames
 - (a) $ROT(\hat{X}, -90)ROT(\hat{Y}, 90)TRANS(6, 0, 2)[T]$
 - (b) $[T]TRANS(6, 0, 2)ROT(\hat{Y}, 90)ROT(\hat{X}, -90)$
 - (c) $[T]ROT(\hat{Y}, 90)TRANS(6, 0, 2)ROT(\hat{X}, -90)$
5. Let us assume that the rotation ${}^U_B R$ given in exercise problem 3 has been represented using the concept of Euler angles. Determine the corresponding values of Euler angles.
6. What do you mean by singularity of a manipulator? How can we check it using the concept of Jacobian matrix?
7. Fig. 2.18 displays a few 2 dof manipulators.
 - (i) Draw their kinematic diagrams.
 - (ii) Assign coordinate systems at various joints of these manipulators according to the rules of D-H parameters setting.
 - (iii) Determine ${}^{Base}_{End-effector} T$.
 - (iv) Draw their workspaces.

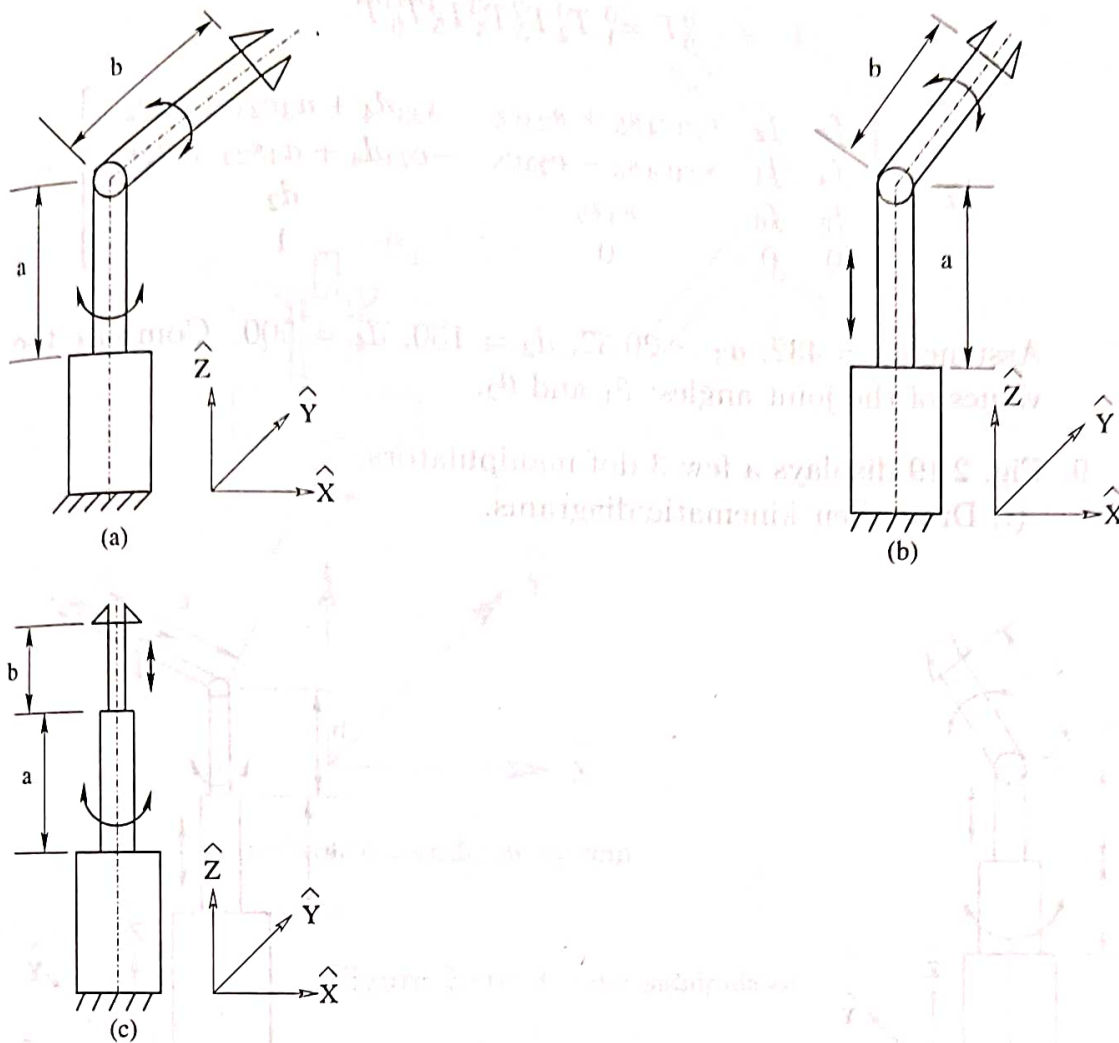


Figure 2.18: Schematic views of a few 2 dof manipulators: (a) TR, (b) SR, (c) TS.

8. Let us suppose that the position and orientation of end-effector of a 6 dof manipulator with respect to its base coordinate system are known, as given below.

$${}^0_6T = \begin{bmatrix} 0 & -1 & 0 & -149.09 \\ 0 & 0 & 1 & 921.12 \\ -1 & 0 & 0 & 20.32 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The following information are given for the manipulator:

$${}^0_1T = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_5T {}^5_6T$$

$${}^1_6T = \begin{bmatrix} f_1 & f_2 & c_{23}c_4s_5 + s_{23}c_5 & s_{23}d_4 + a_3c_{23} + a_2c_2 \\ f_3 & f_4 & s_{23}c_4s_5 - c_{23}c_5 & -c_{23}d_4 + a_3s_{23} + a_2s_2 \\ f_5 & f_6 & s_4s_5 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Assume $a_2 = 432$; $a_3 = 20.32$; $d_2 = 150$; $d_4 = 500$. Compute the values of the joint angles: θ_1 and θ_2 .

9. Fig. 2.19 displays a few 3 dof manipulators.

(i) Draw their kinematic diagrams.

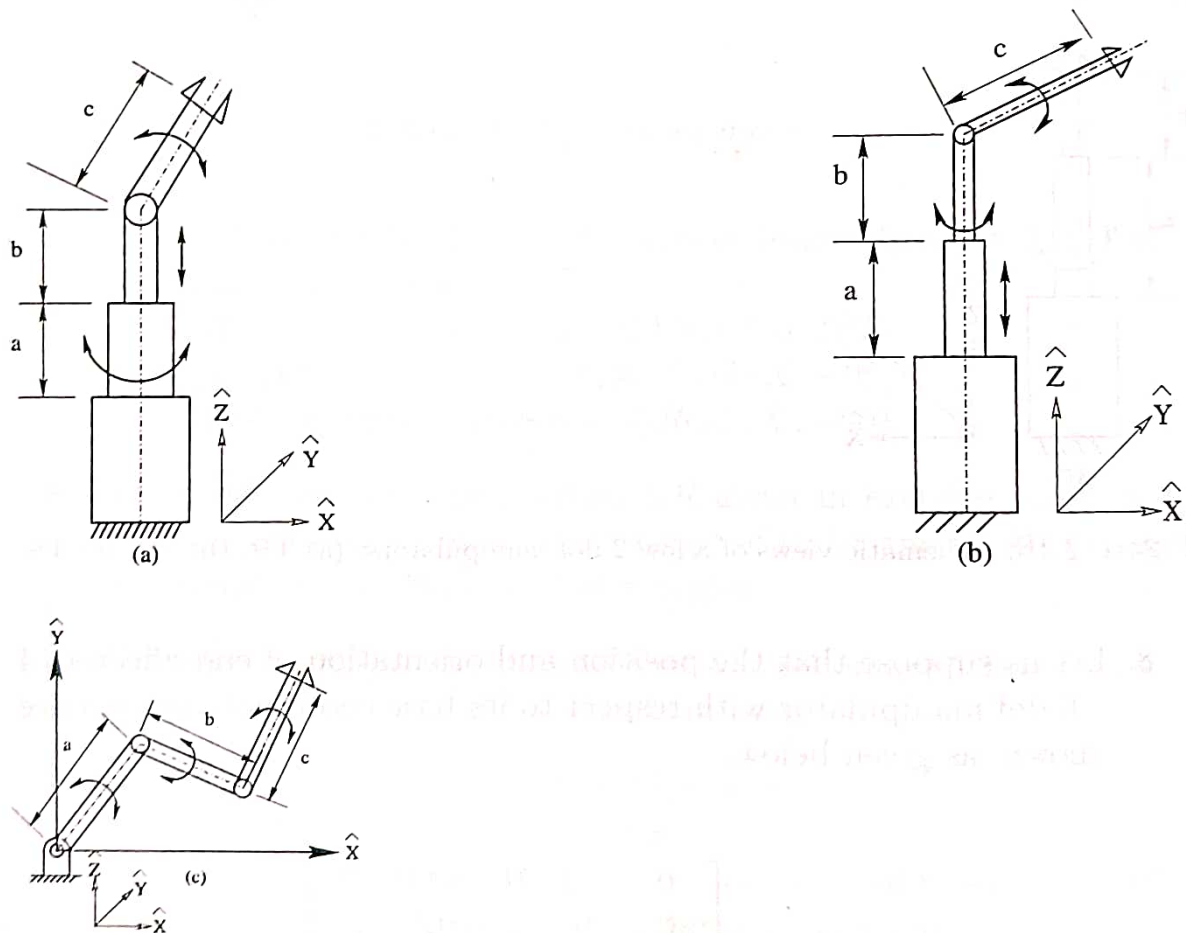


Figure 2.19: Schematic views of a few 3 dof manipulators: (a) TSR, (b) STR, (c) RRR.

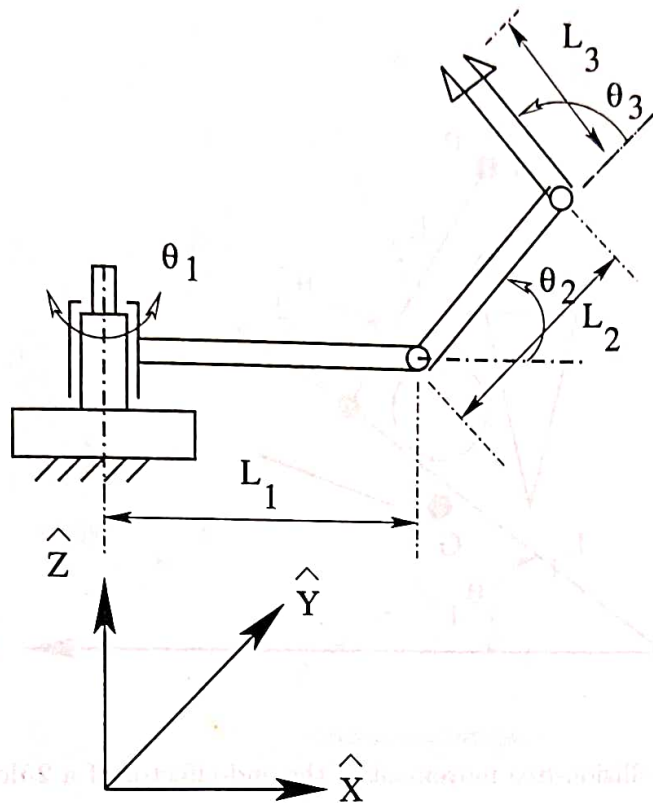
(ii) Assign coordinate systems at various joints of these manipulators according to the rules of D-H parameters setting.

(iii) Determine ${}^{Base}_{End-effector}T$.

(iv) Draw their workspaces.

10. Fig. 2.20 shows the sketch of a manipulator.

(i) Establish link coordinate systems at the different joints.



reference coordinate system

Figure 2.20: A 3 dof manipulator.

- (ii) Determine ${}^{Base}_{End-effector}T$.
 - (iii) Carry out its inverse kinematics.
 - (iv) Draw its workspace.
11. Let us consider a 2 dof manipulator working in 2-D plane as shown in Fig. 2.21. The lengths of two links are denoted by L_1 and L_2 , and L_1 is assumed to be more than L_2 . Starting from an initial position S , the tip of the manipulator P will have to reach the final position G after avoiding collision with the stationary point, line, triangular and circular obstacles, as shown in the said figure. Draw the plots of θ_1 vs. θ_2 to ensure collision-free movement of the tip. Use the following data: $L_1 = 15cm$; $L_2 = 12cm$; $S : (2.0, 5.0)$; $G : (2.5, 1.5)$; Circular obstacle: center $(2.5, 2.5)$ and radius $= 1.0cm$; Point obstacle: $(2.0, 4.0)$; Line obstacle: from point $(3.0, 1.5)$ to point $(4.5, 2.0)$; Triangular obstacle: coordinates of vertices $(1.5, 1.5)$, $(1.0, 3.5)$ and $(2.0, 3.5)$.

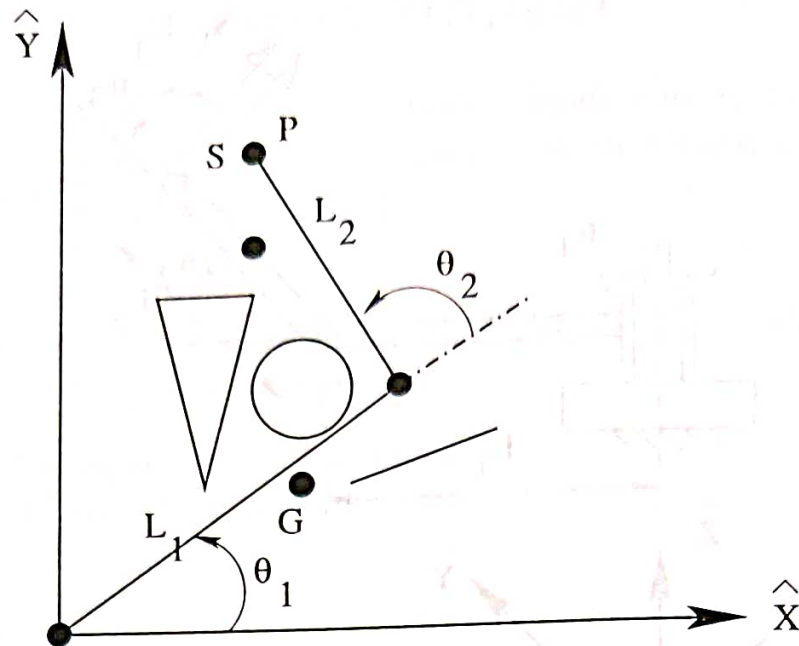


Figure 2.21: Collision-free movement of the end-effector of a 2 dof manipulator.

12. Define various link and joint parameters used in D-H parameters' setting at different joints of a manipulator.
13. How can we relate Cartesian velocity of a manipulator with its joint velocity? Explain.
14. How can we use the concept of Jacobian matrix to control a manipulator? Explain.
15. Why do the industrial robots have 6 dof in general? Explain.
16. How can the position and orientation of a body be represented in different systems? Explain.
17. Discuss the properties of translation and rotation operators.
18. Explain the principle of composite rotation matrix.
19. Define the problems of forward and inverse kinematics of a manipulator.
20. Fig. 2.22 shows a 6 dof manipulator.
 - (i) Draw its kinematic diagram.
 - (ii) Establish coordinate systems at various joints of this manipulator.
 - (iii) Prepare D-H parameters' table.
 - (iv) Determine ${}^{\text{Base}}T_6$.

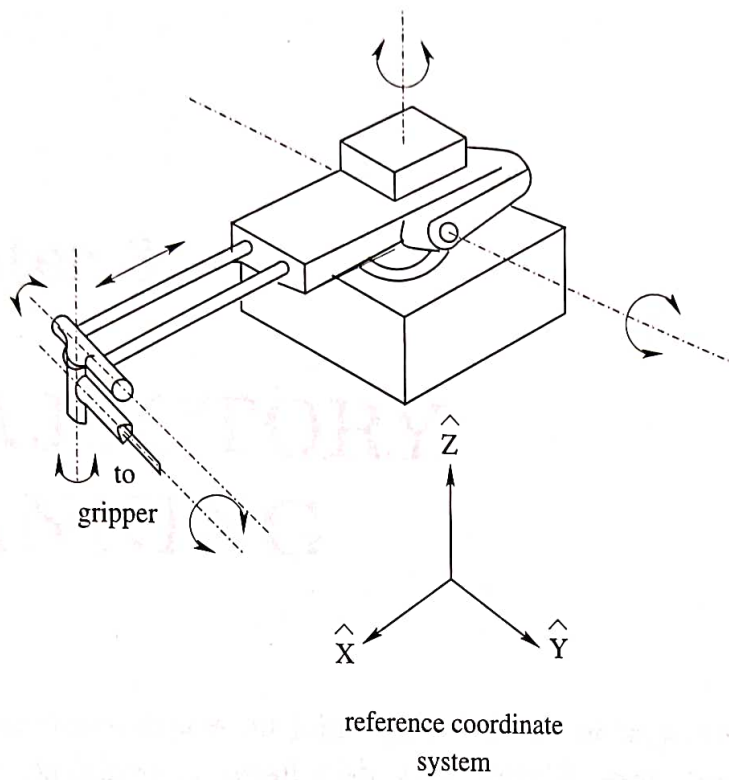


Figure 2.22: A 6 dof manipulator.

- (v) Draw its workspace.
- (vi) Carry out its inverse kinematics.